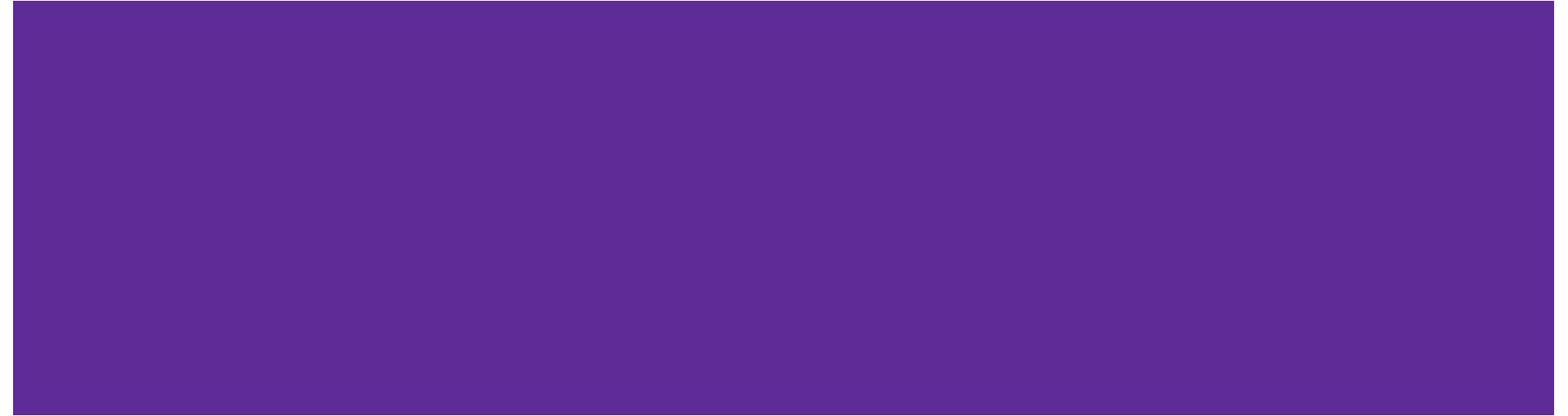


Machine Learning CA3

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Fruit Image Classification Using Deep Learning Models



Fruit Image Classification

Our aim is to classify images of 5 fruits classes: apples, bananas, grapes, mangoes, strawberries

Compare and contrast the performance of four models:

- Simple CNN, Enhanced CNN
- Two Advanced Deep Learning models

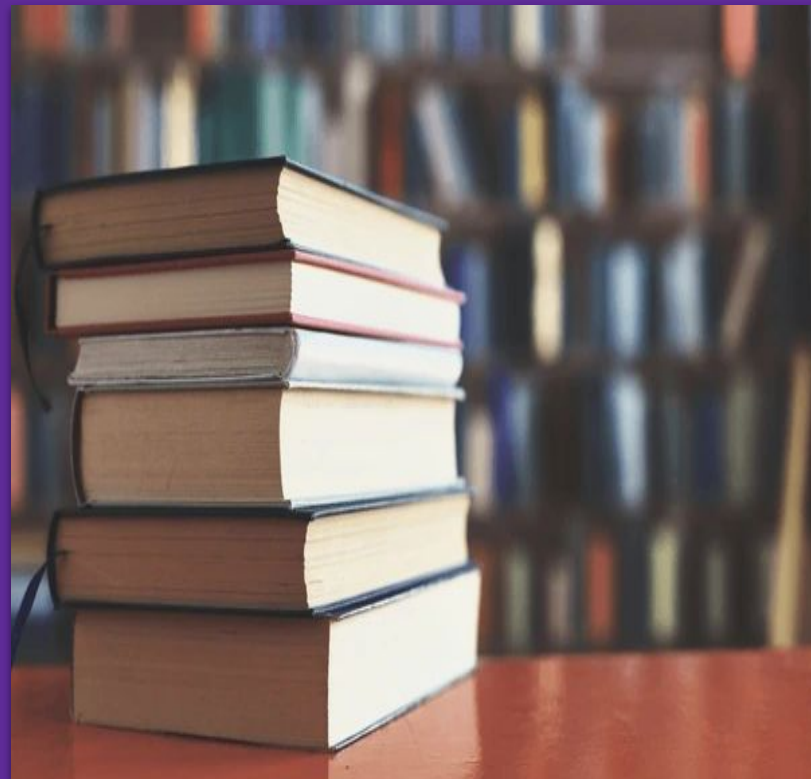
Evaluate accuracy and performance

Identify strengths and limitations of each model

Literature Review

- Krizhevsky et al. showed that deeper CNN architectures can capture complex visual patterns
- Kumala et al. compared MobileNetV2 and VGG16 and found that MobileNetV2 achieved higher accuracy whilst being more efficient

This literature supports the comparison of custom CNNs with TL models in this task.

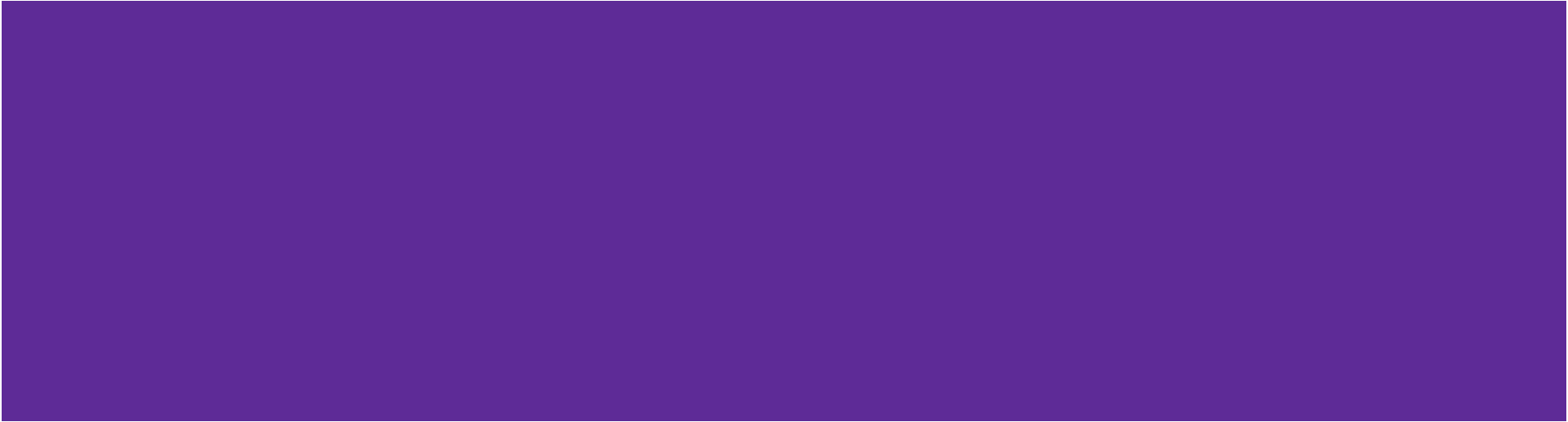


[Image Source](#)

Problem Definition

- Multi-class image classification (5 fruits: apples, bananas, grapes, mangoes, strawberries)
- Challenge: variations in lighting, angle, size, background, and similar-looking classes
- Compare model performance on the same task:
 - Custom CNNs (baseline)
 - Advanced Deep Learning Models: MobileNetV2, VGG16
- Evaluation: accuracy, precision, recall, F1-score, confusion matrix

Data Exploration and Understanding



Dataset Overview

Kaggle Fruit Image Dataset (10,000 images)

5 classes: apples, bananas, grapes, mangoes, strawberries

Organised into train/test folders by class

Automatically labelled for TensorFlow use

Train/test split enables evaluation on unseen data

EDA

Exploratory Data Analysis

- Verified dataset structure and labels
- Checked image sizes, consistency, and overall dataset size
- Visually inspected sample images
- Images resized to 128×128 and batched (32) for training

Key observations:

- Varied lighting, angles, and image conditions
- Balanced classes across all fruit types
- No missing or corrupted data

Data Pre-processing and Cleaning

- Images were resized to 128×128 to ensure consistency across all images and also for efficiency
- Pixel values were normalised to $[0 \text{ to } 1]$ from the range $(0 \text{ to } 255)$ using a recycling factor of $1/255$ for stable training
- Data was processed in batches of 32 to further aid in performance
- Dataset was split into training and test sets for evaluation of the unseen data

Model Implementation



Baseline CNN

- 2 Convolution layers (32 & 64 filters, ReLU)
- Max pooling after each convolutional layer
- Flatten layer
- Dense layer (64 neurons, ReLU)
- Output layer (5 classes, Softmax)
- Optimizer: Adam
- Loss function: Sparse Categorical Cross-Entropy
- Trained on 0.001 learning rate

Extended CNN

- 3 Convolution layers (32, 64, 128 filters)
- Max pooling after each convolutional layer
- Flatten layer
- Dense layer
- Output layer (5 classes)
- Additional Convolutional layer added depth to capture more complex image features
- Trained on 0.001 learning rate

MobileNetV2

- The original top classification layers were removed
- The pre-trained base was frozen
- A new classification head was added consisting of a Global Average Pooling layer, a Dropout layer and a Dense output layer with 5 neurons and softmax activation

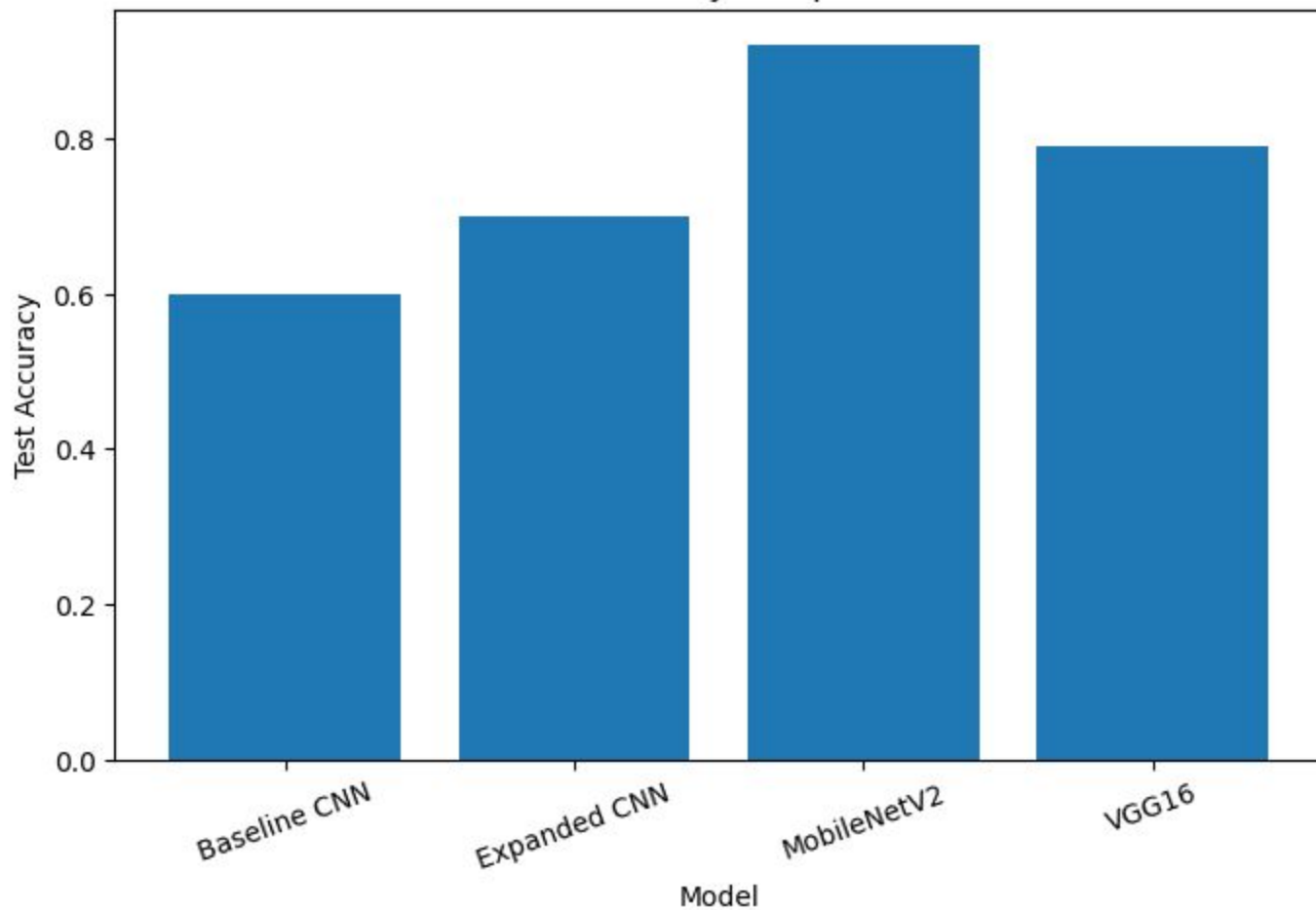
VGG16

- The original top classification layers were removed
- The convolutional base was frozen
- New task-specific layers were added: a Flatten layer, two Dense layers with ReLU activation and a final dense output layer with 5 neurons and softmax activation

Model Evaluation



Test Accuracy Comparison



Test Accuracy Results

- Baseline CNN: 60%
- Expanded CNN: 70%
- VGG16: 79%
- MobileNetV2: 92%

Apples were the most difficult class across multiple models, suggesting greater visual overlap with other fruit categories.

Critical Analysis and Interpretation

- The expanded CNN improved over the baseline CNN -> extra layers helped capture more complex patterns
- The TL models outperformed the custom CNNs -> they reused useful features learned from ImageNet
- MobileNetV2 performed better than VGG16 -> a larger model does not always guarantee better results

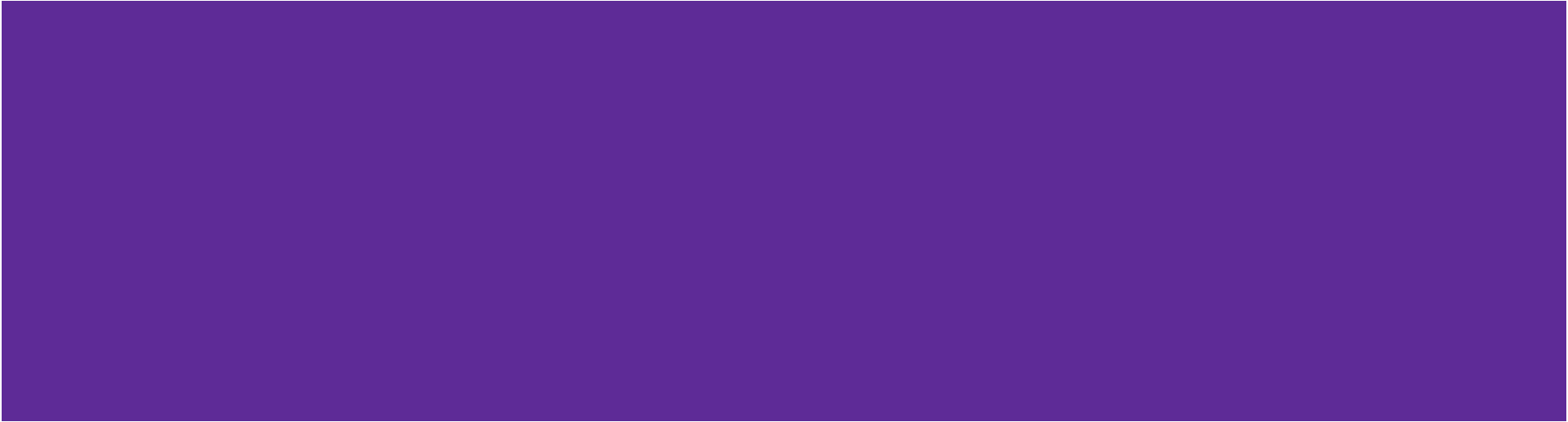
A key limitation: the dataset is relatively balanced and controlled, so real-world performance may drop when images vary more.

Conclusion



- This project compared four DL models for classifying fruit images into five categories
- TL clearly produced the **strongest results**
- MobileNetV2 was the best-performing model with **92% accuracy**
- The custom CNN models were **useful baselines**, but they were outperformed by the pre-trained models
- Future improvements could include data augmentation, fine-tuning more layers, and testing on more varied real-world fruit images

AI Fairness Evaluation Assignment



Part A Findings

Dataset 1: Full-size images performed best overall:

- Best resize level: 100%
- Resizing reduced structural quality, especially SSIM

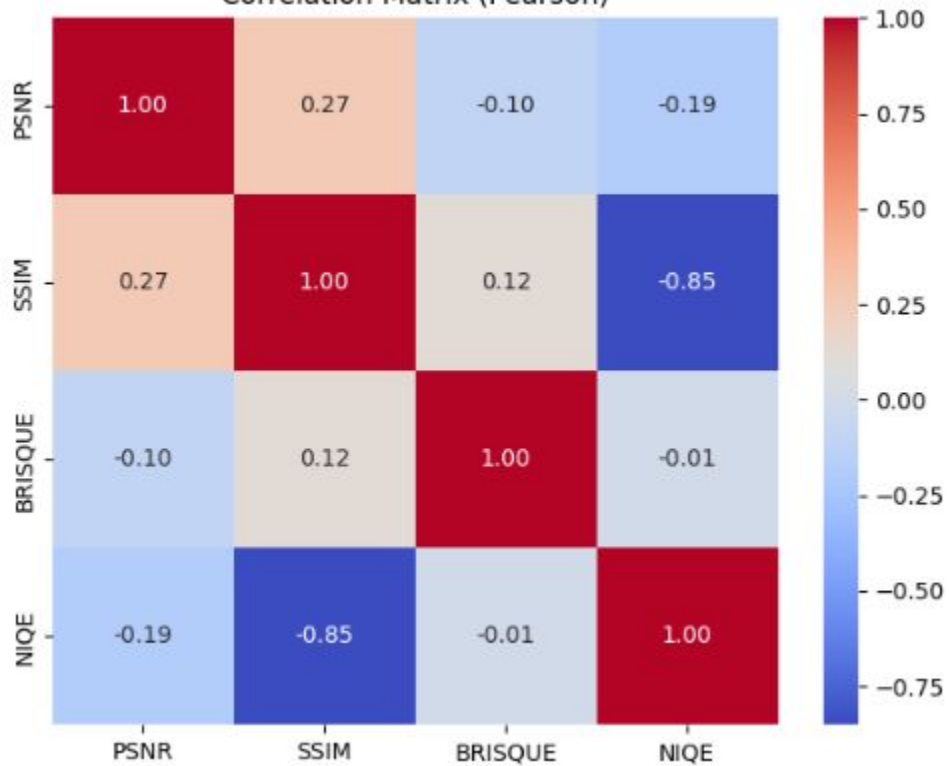
Dataset 2: Results showed a trade-off:

- 100% was best for SSIM and NIQE
- 50% gave the best BRISQUE result -> moderate resizing may reduce some noise/artefacts

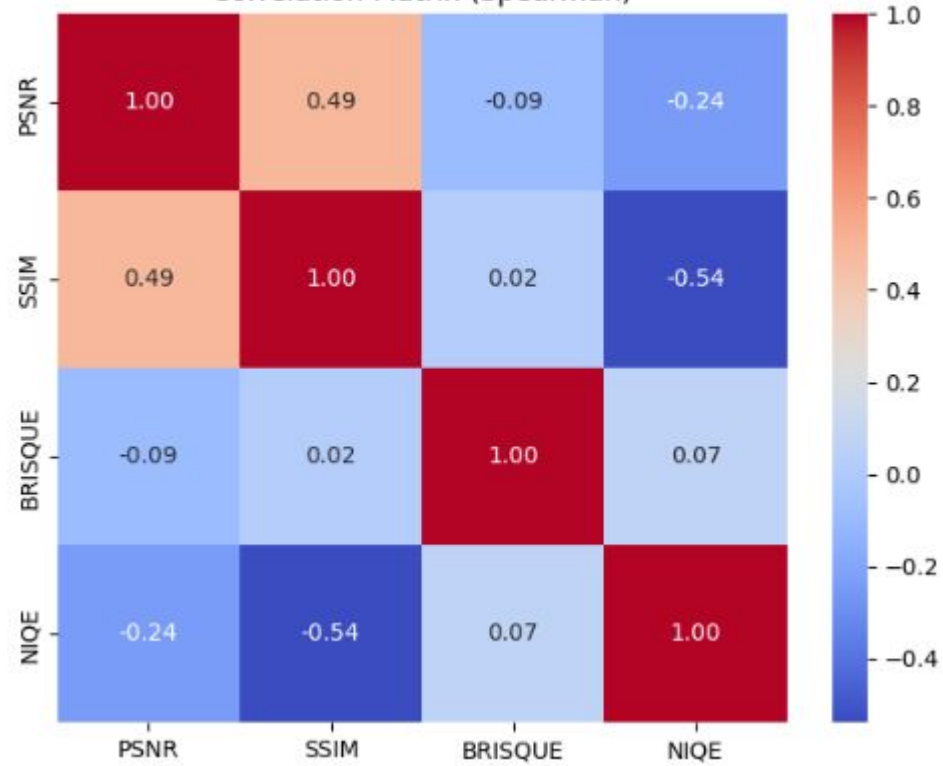
Conclusion

Overall, keeping images at **full size gave the most reliable quality**, especially when preserving image structure was important. Although, **50% resizing may be acceptable for Dataset 2** when speed or BRISQUE quality is prioritised.

Correlation Matrix (Pearson)



Correlation Matrix (Spearman)



Part B Findings

Metric Correlations:

- Strongest relationship: SSIM and NIQE (-0.85) both move together across datasets
- PSNR and SSIM show a slight agreement of 0.27 in pearson and 0.49 in the spearman overall but diverge strongly in dataset 1 being low light conditions
- BRISQUE correlates weakly against the other metrics as it operates independently capturing perceptual quality the other metrics miss

Computational Timing:

- Processing time scales with resolution, Dataset 2 ranged from 5.9 ms (25%) to 56.8 ms (100%)
- 50% resize gives the best trade off at 4-5x speedup with minimal quality loss

Scaling Factor Use in Underwater Imagery:

- 25% - good for simple tasks such as avoiding large obstacles (ridges, trenches etc)
- 50% - identifying large marine life, live streaming
- 75% - suited for upclose inspection of marine life such as coral reefs and habitats

Part B Findings

Conclusion

Overall based on the experiments conducted, these are the following recommendations:

Always pair a reference based metric (PSNR/SSIM) with a reference-less metric (NIQE/BRISQUE)

No single metric tells the full story.

Avoid using BRISQUE on its own due to its low correlation with PSNR and SSIM

Be mindful that Metrics differ depending on the imagery domain

50% resize is the recommended default as it offers a strong speed-quality trade-off

For more detail critical tasks: use 75% - 100%

Thank you

References

- Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. In Advances in neural information processing systems (Vol. 25).
<https://papers.nips.cc/paper/4824-imagenet-classification-with-deep-convolutional-neural-networks>.
- Kumala, R. A., Sari, C. A., & Rachmawanto, E. H. (2025). A comparison of mobilenetv2 and vgg16 architectures with transfer learning for multi-class image-based waste 10 classification. Journal of Applied Informatics and Computing, 9 (4), 1610–1624.
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